



UNIVERSIDAD TÉCNICA ESTATAL DE QUEVEDO
FACULTAD DE CIENCIAS COMPUTACIONALES Y DISEÑO DIGITAL
INGENIERÍA TELEMÁTICA



ASIGNATURA:

ANTENAS Y PROPAGACIÓN

DOCENTE:

ING. TORRES QUIJIJE ANGEL IVAN

ALUMNO:

MORALES COBEÑA MIYAKO KUSHIRO

DÉCIMO SEMESTRE

2025-2026 PPA

QUEVEDO, ECUADOR

9 DE AGO. DE 25

1. PERFIL DEL TRAYECTO

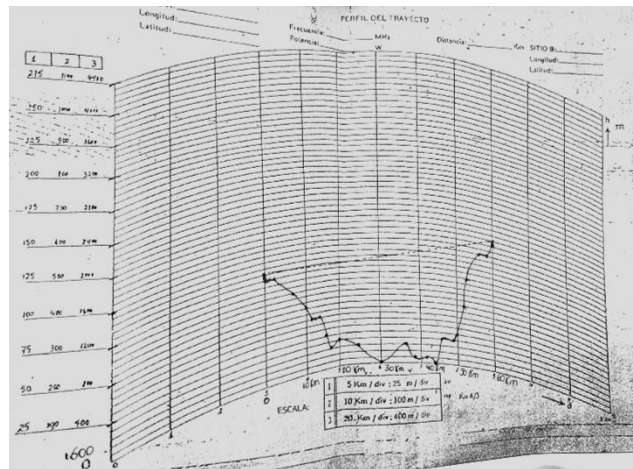


Ilustración 1. Perfil del Trayecto.

2. SIMULACIÓN UBIQUITI

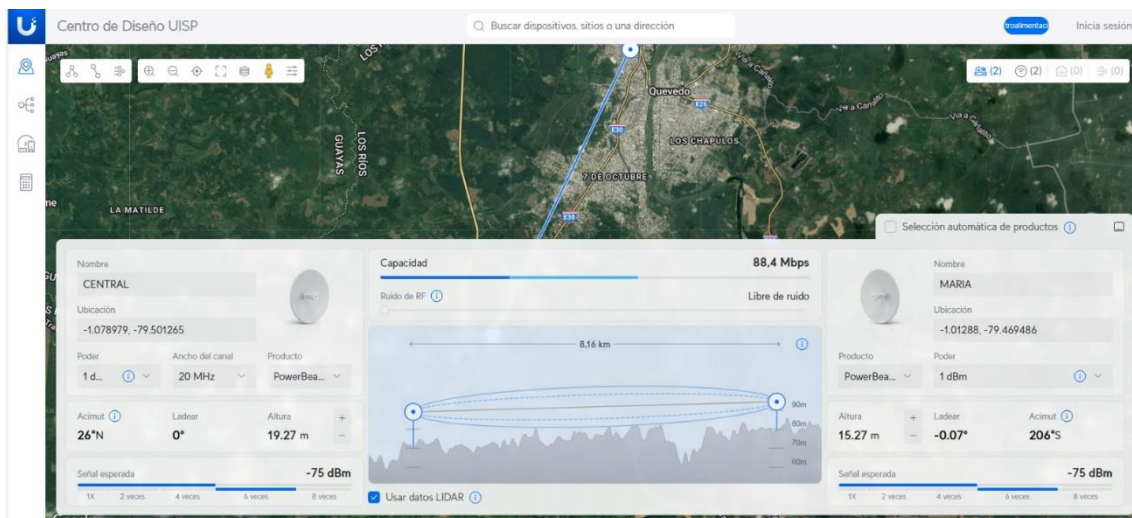


Ilustración 2. Ubiquiti.

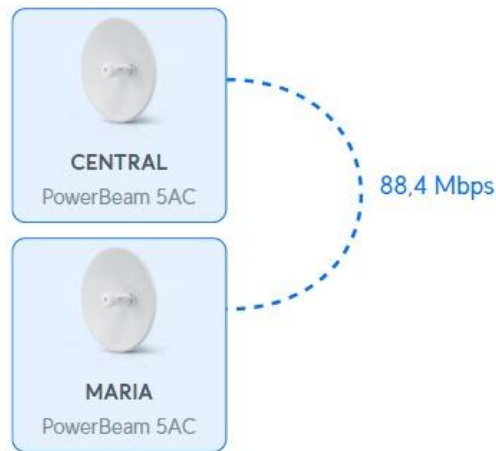


Ilustración 3. Capacidad.

Lista de dispositivos Restablecer valores					Total de elementos seleccionados \$238.00	
☒	Inalámbrico	Estado	Cantidad	Precio Por Unidad	Precio	IVA no incluido
☒	PowerBeam 5AC	En stock	2	\$119.00	\$238.00	
					\$238.00	

Caja

Total **\$238.00**

Ilustración 4. Lista de dispositivos.

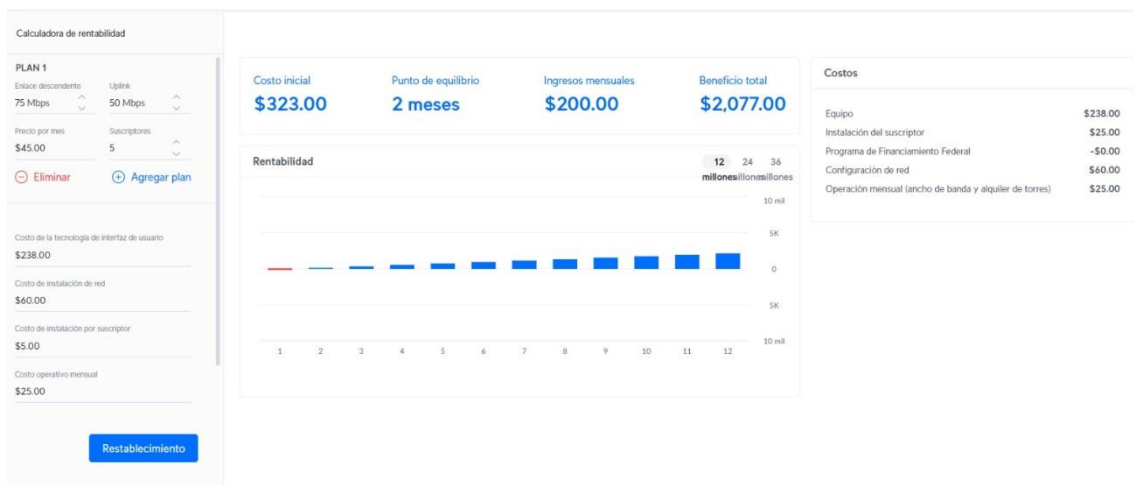


Ilustración 5. Rentabilidad.

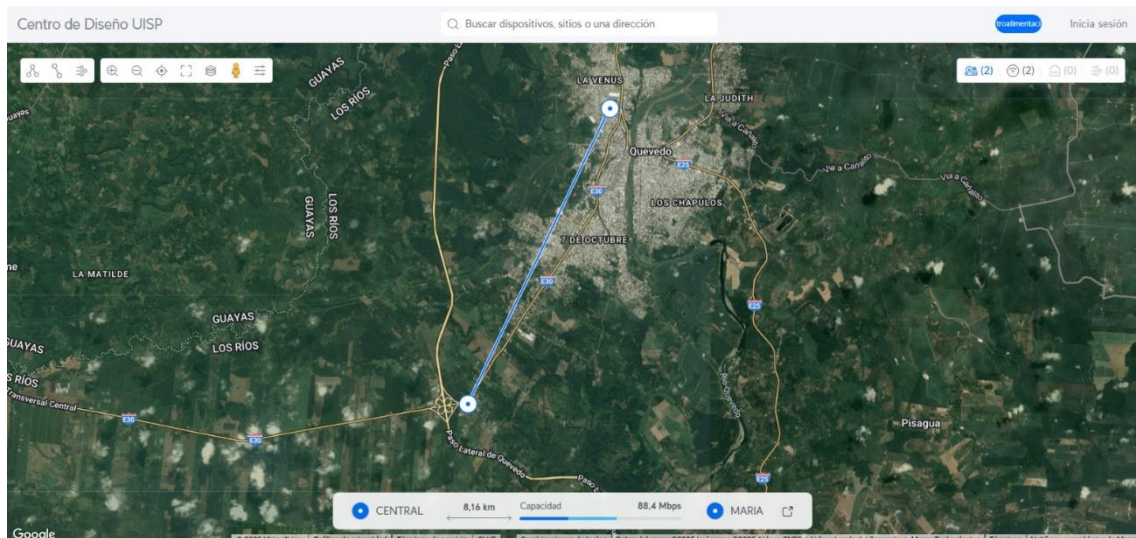


Ilustración 6. Enlace.

3. SIMULACIÓN RADIO MOBILE DEL CAMPUS CENTRAL DE LA UTEQ A LA UNIÓN

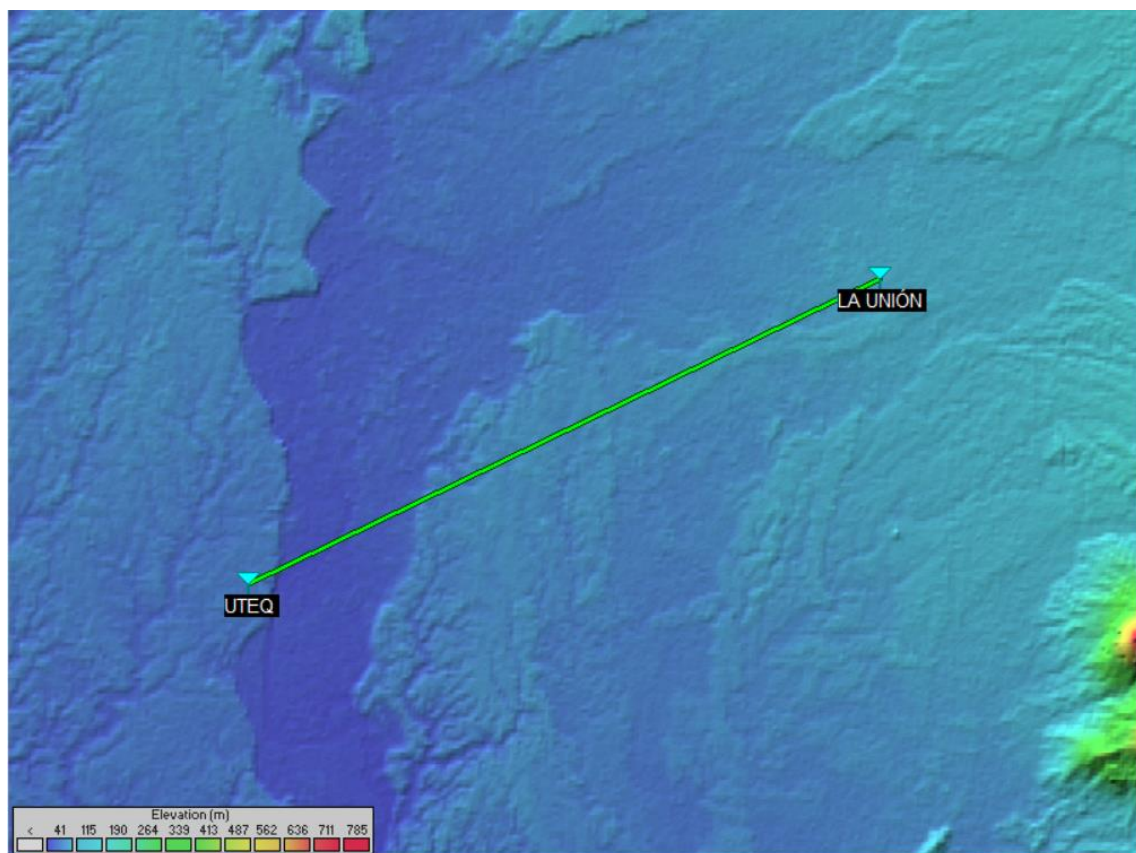


Ilustración 7. Enlace.

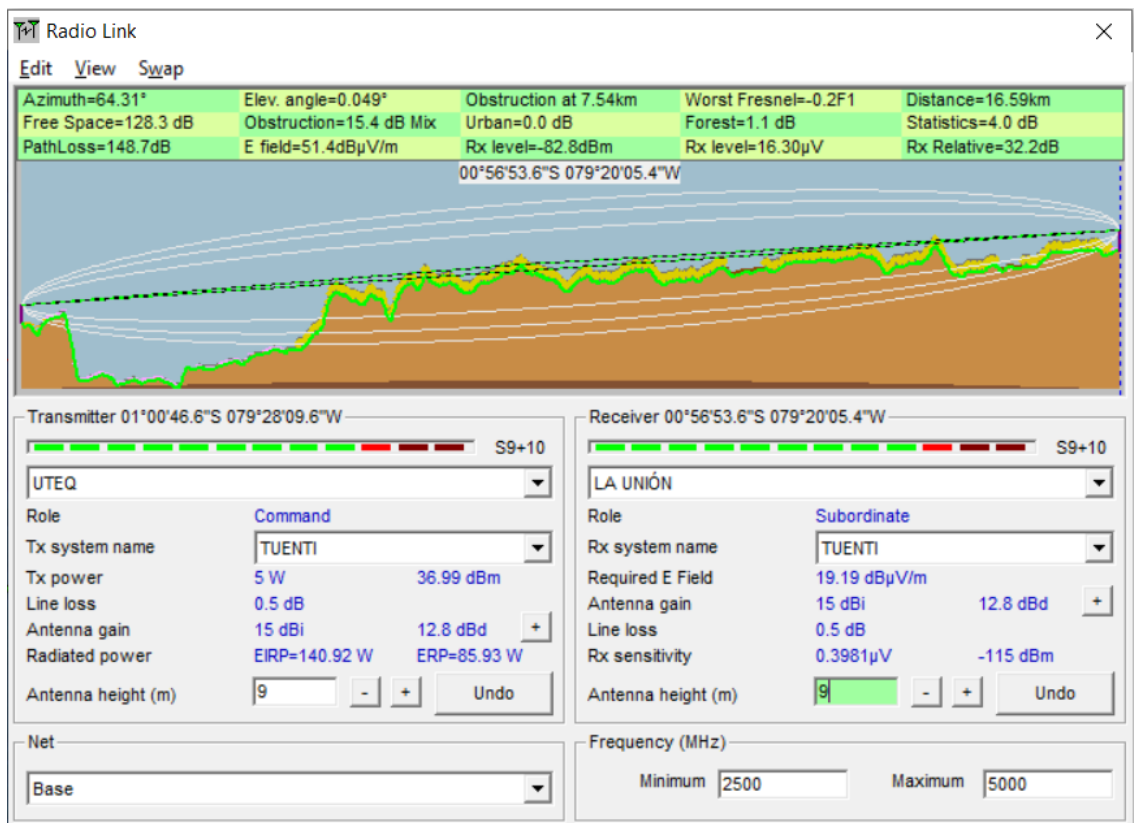


Ilustración 8. Radio Link

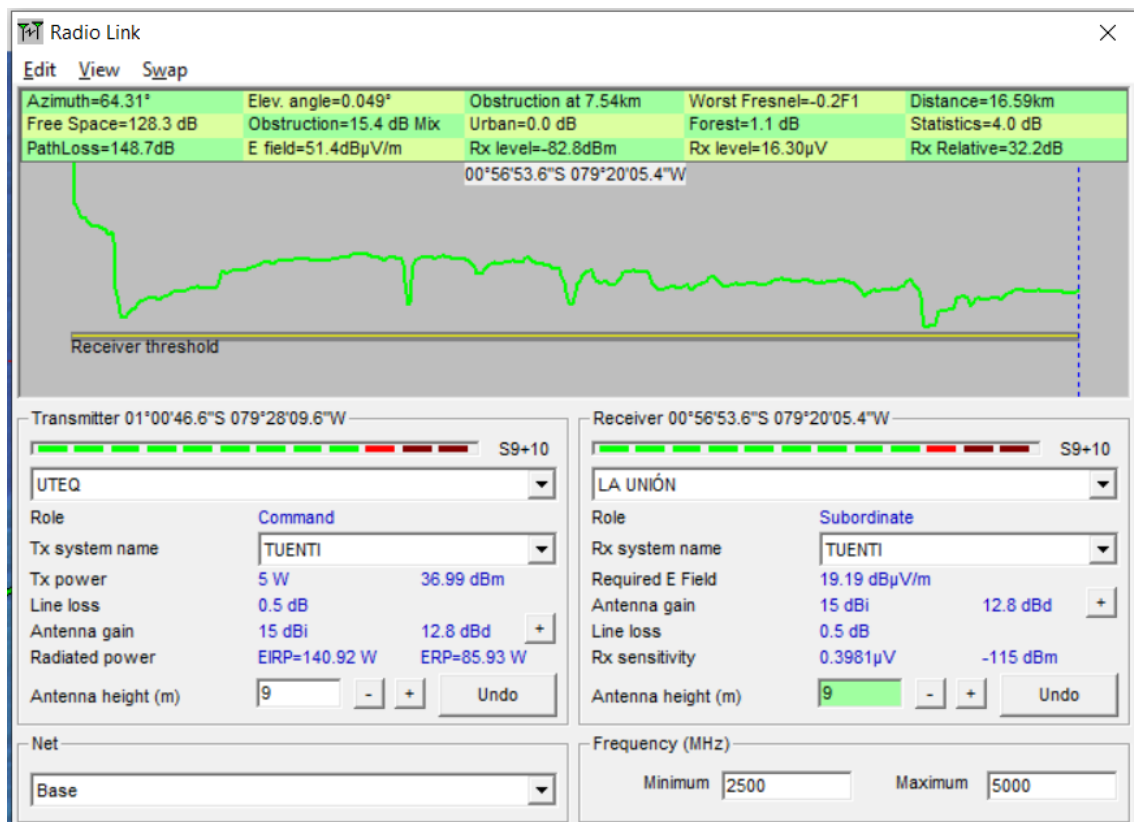


Ilustración 9. Range.

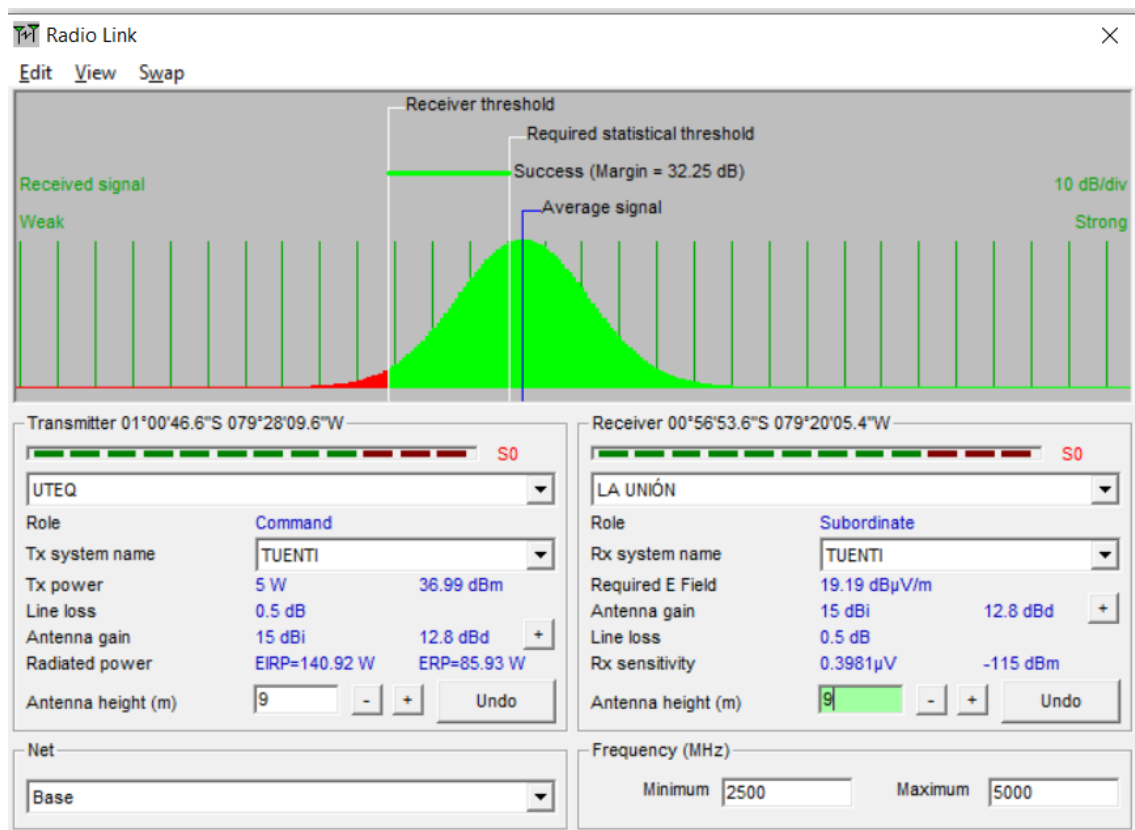


Ilustración 10. Distribución.

Distance between UTEQ and LA UNIÓN is 16.6 km (10.3 miles)
 True North Azimuth = 64.31°, Magnetic North Azimuth = 67.43°, Elevation angle = 0.0489°
 Terrain elevation variation is 71.0 m
 Propagation mode is diffraction, single obstruction, 0.2F1 at 7.5km
 Average frequency is 3750.000 MHz
 Free Space = 128.3 dB, Obstruction = 15.4 dB Mix, Urban = 0.0 dB, Forest = 1.1 dB, Statistics = 4.0 dB
 Total propagation loss is 148.7 dB
 System gain from UTEQ to LA UNIÓN is 181.0 dB (dipole.ant at 64.3 °0.05° gain = 15.0 dBi)
 System gain from LA UNIÓN to UTEQ is 181.0 dB (dipole.ant at 244.3 °-0.20° gain = 15.0 dBi)
 Worst reception is 32.2 dB over the required signal to meet
 50.000% of time, 50.000% of locations, 70.000% of situations

Ilustración 11. Detalles.

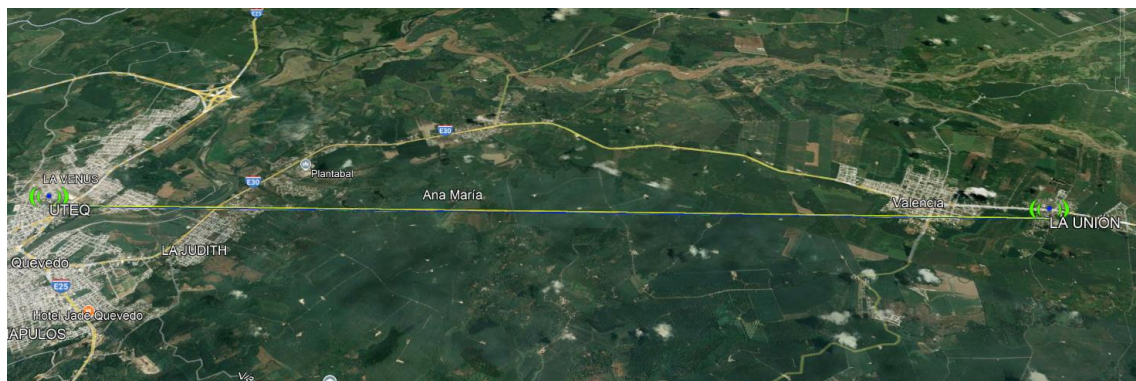


Ilustración 12. Google Earth.